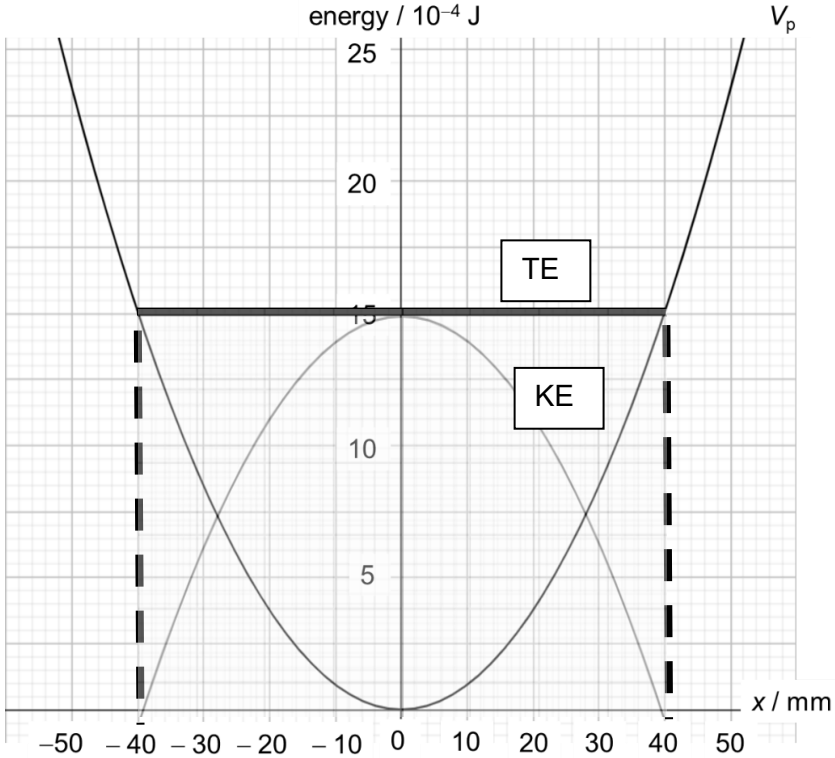
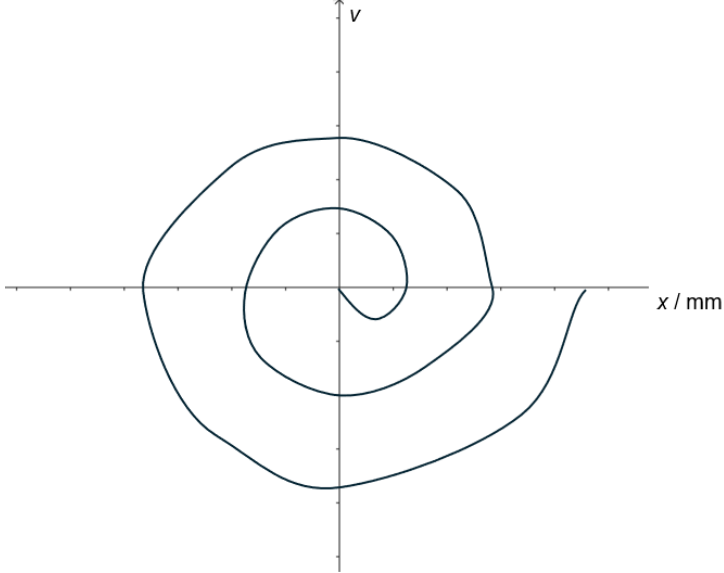


Questions			Answers	Marks
2	(a)		Its <u>acceleration is directly proportional to its displacement</u> from a fixed point (equilibrium position) and is always <u>directed towards that fixed point</u> .	B1
	(b)		gain in gPE = mgh	C1
			$(0.150)(9.81)(1.0 \times 10^{-3}) = 1.4715 \times 10^{-3} \sim 1.5 \times 10^{-3} \text{ J}$	A1

(c)	(i) (ii)		B1 B2
(d)		From calculation in (b), total energy is $1.5 \times 10^{-3} \text{ J}$ So the horizontal displacement should be 40 mm	B1
(e)		 Direction of displacement must be opposite to that of velocity, i.e. if line is in positive displacement, velocity must be in negative region Subsequent quarter of oscillation must decrease in amplitude and magnitude of velocity.	B2

